

## Project Title:

### Encrypt S3 Bucket using AWS KMS (Customer Managed Key)

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## Project Description:

Implemented secure data storage in Amazon S3 by enabling **server-side encryption using AWS Key Management Service (KMS)**. A **Customer Managed Key (CMK)** was created and integrated with the S3 bucket to ensure that all uploaded objects are encrypted automatically.

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## Objective:

- To secure S3 data using **SSE-KMS**
  - To understand **AWS KMS key creation and management**
  - To enforce **default encryption at the bucket level**
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## Services Used:

- **Amazon S3**
  - **AWS KMS (Key Management Service)**
  - **IAM (Identity and Access Management)**
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## Implementation Steps:

### 1. Created S3 Bucket

- Navigated to S3 service
  - Created a new bucket with a unique name
  - Selected region (ap-south-1)
  - Kept default configurations
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### 2. Created Customer Managed Key (CMK)

- Opened AWS KMS console

- Created a new key with:
    - Key type: Symmetric
    - Usage: Encrypt and Decrypt
  - Assigned alias (s3-kms-key)
  - Defined:
    - Key administrators
    - Key usage permissions
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### 3. Configured Default Encryption

- Opened S3 bucket properties
  - Enabled **Default Encryption**
  - Selected:
    - **SSE-KMS (AWS Key Management Service)**
  - Chose the created CMK (s3-kms-key)
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### 4. Uploaded Object to S3

- Uploaded a test file into the bucket
  - Ensured automatic encryption was applied
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### 5. Verified Encryption

- Opened object properties
  - Confirmed:
    - Encryption type: **AWS-KMS**
    - Key used: **Customer Managed Key**
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**Architecture:**

User Upload → S3 Bucket → SSE-KMS Encryption → AWS KMS (CMK)

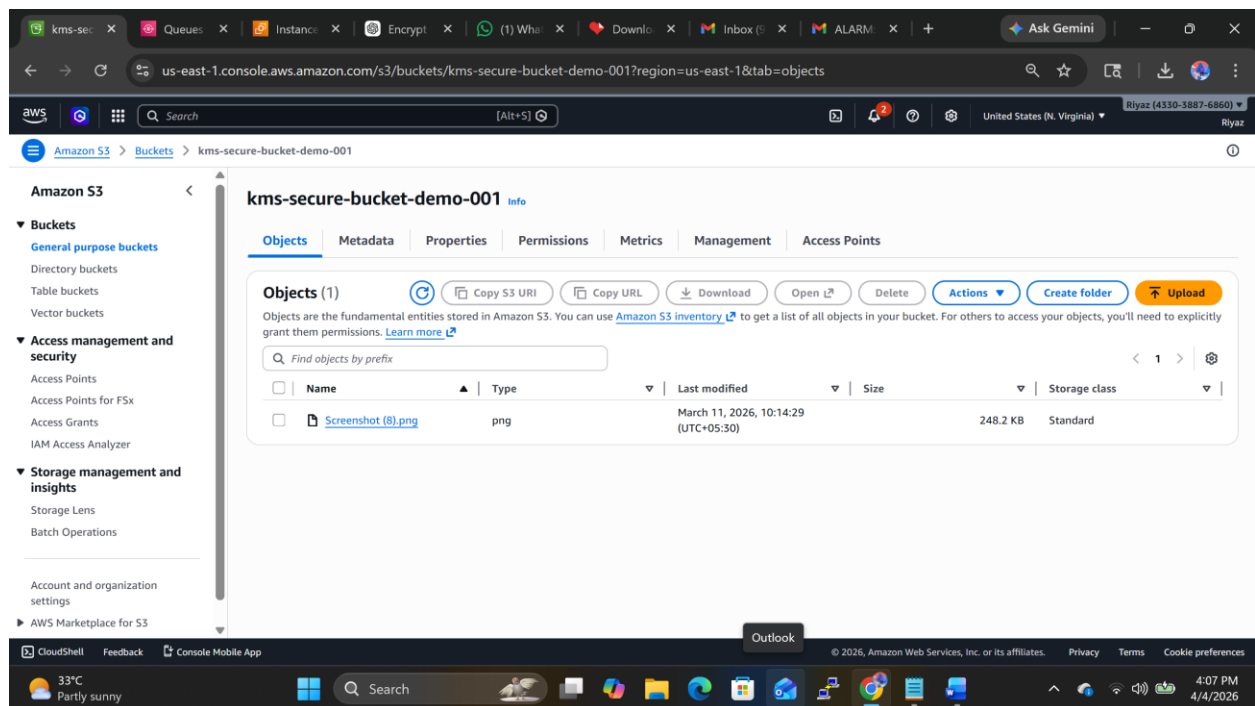
## Outcome / Result:

- Successfully configured **automatic encryption** for S3 objects
- Ensured **data security using KMS-managed keys**
- Verified encryption at object level

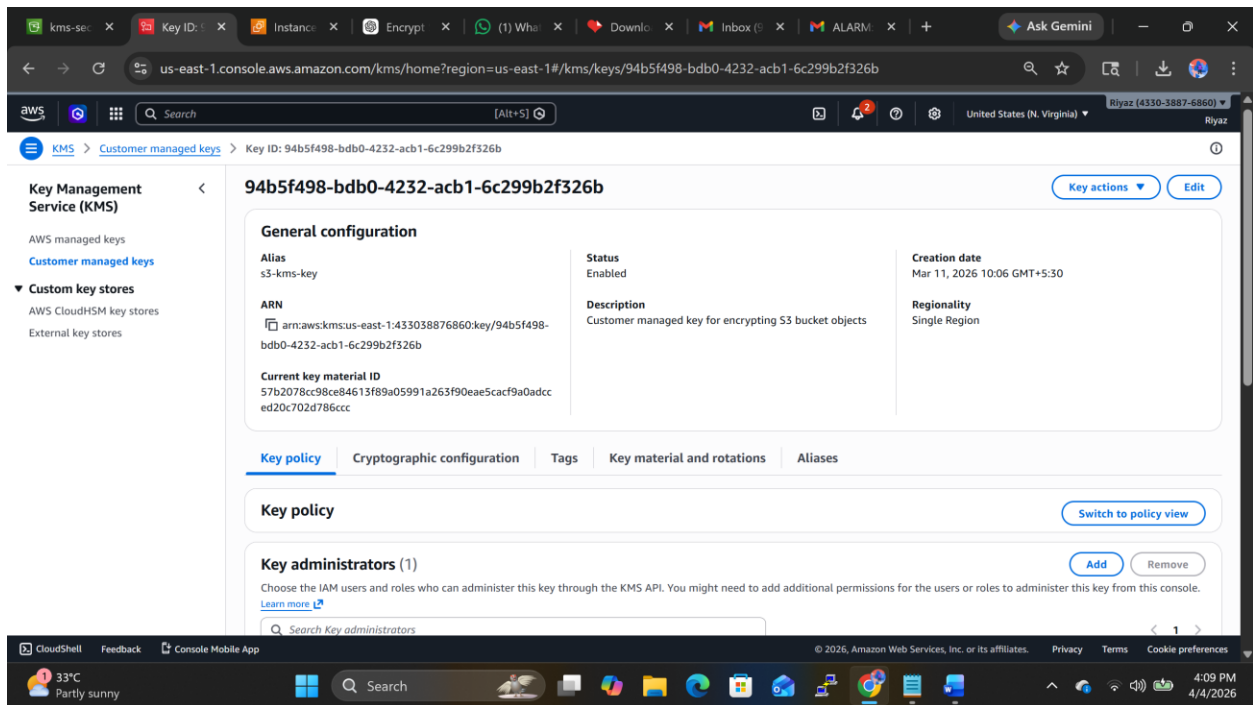
## Key Learnings:

- Difference between **AWS Managed Keys vs Customer Managed Keys**
- How **SSE-KMS works internally with S3**
- Importance of **encryption for cloud security and compliance**

S3 bucket created:



KMS key created:



Files upload using encryption key:

